

SHIVAM KUMAR SAH

Machine Learning & AI Engineering Aspirant | NLP, RAG & Applied Deep Learning

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Machine learning student with hands-on, project-proven experience across NLP, retrieval-augmented generation (RAG), and applied deep learning, focused on building socially relevant AI systems for real-world constraints — low-resource languages, offline-first environments, and regulated domains like healthcare and banking. Shipped end-to-end work spanning CNN-based medical image classification (EfficientNet-B4, ResNet), multilingual/code-switched speech systems, and clinical-compliance platforms for hackathons. Comfortable owning a project from architecture and research through to a working prototype.

EDUCATION

DIT University — B.Tech in Computer Science Engineering (AI/ML) | 2023–2027 | CGPA: 7.80/10

Uniglobe High School — Science Stream | 2021–2023 | GPA: 3.24/4.0

TECHNICAL SKILLS

Programming: C, Java, Python, Data Structures & Algorithms, HTML, CSS, MySQL

Machine Learning & AI: Deep Learning, NLP, Retrieval-Augmented Generation (RAG), LLMs, Computer Vision, NumPy, Pandas, Scikit-Learn, PyTorch, NLTK

AI Tools & Ecosystem: LLM prompt engineering, RAG pipeline design, transfer learning / fine-tuning (EfficientNet, ResNet), XLM-RoBERTa

Cloud & IoT: AWS, Firebase, microcontrollers, sensors, automation systems

Dev Tools: Git/GitHub, Jupyter, VS Code, google colab, Jupyter notebook

PROJECTS

Nepali-English Code-Switch Language Identifier: Built a token-level language identification model (XLM-RoBERTa) for Nepali-English code-switched text, achieving 99.5% accuracy on a self-annotated 121K-token dataset..

Multilingual Voice Banking Sahayak (MVBS) — Low-Resource Nepali Voice Banking System(**Pending**)

- Architected and authored a full bank-grade technical research paper for a voice-banking assistant handling NepGlish (Nepali-English) code-switching.
- Designed the ASR pipeline around Whisper Large-v3 and IndicConformer, with proactive safety checks for destructive banking actions and graceful degradation under low-resource conditions.

Nepal Flood Prediction System — RNN/LSTM Forecasting Model

- Built a six-phase roadmap benchmarking baseline RNN vs. LSTM architectures for flood forecasting using DHM Nepal and NASA POWER API data sources.

Skin Cancer Classification (HAM10000 / Skin Cancer MNIST) — Kaggle

- Trained and optimized an EfficientNet-B4 CNN classifier on the HAM10000 dermatoscopic image dataset, reaching ~75% classification accuracy.

Additional Projects: Image Classification Model (42,000+ images, CNN + augmentation) • Breast Cancer Prediction (feature-based ML classifier) • Doctor-Finding App (healthcare app, backend API) • Smart IoT Systems (Smart Bin & automation) • Portfolio Website

EXPERIENCE

IT Training Nepal Centre — Intern (3 Months)

- Worked on ML tooling, real-world IT projects, and collaborative development environments alongside a technical team.

RESEARCH INTERESTS

- NLP Research — Sentiment analysis, summarization, and conversational AI / chatbots.
- IoT Research — Automation systems and real-time embedded components.

LANGUAGES & INTERESTS

Languages: English (Fluent), Nepali (Native), Hindi (Fluent)

Interests: AI Research, NLP, IoT, Emerging Tech, Photography, Music